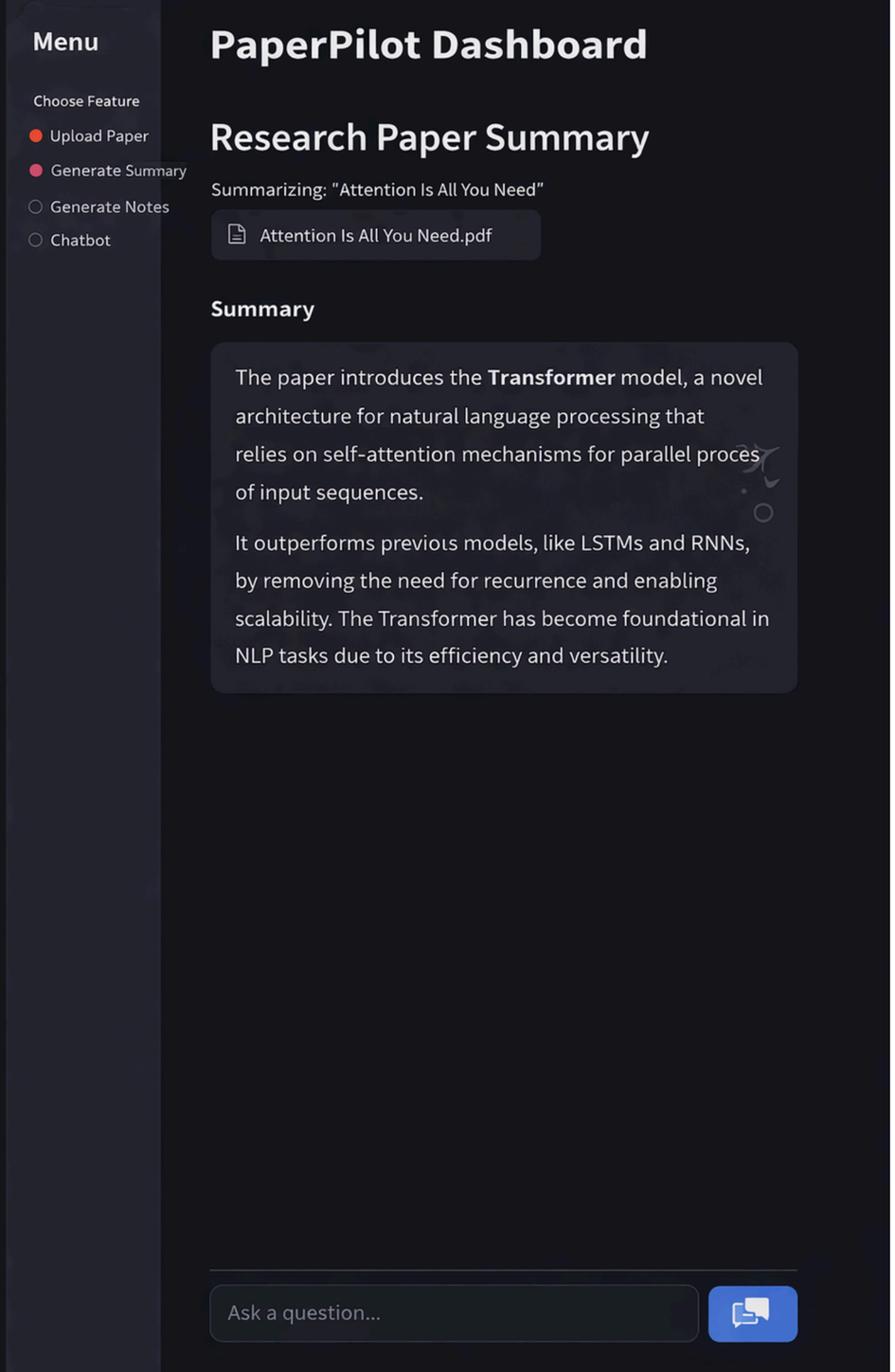


PaperPilot – GenAI Research Assistant

An AI-powered solution for rapid document analysis and question-answering in academic and professional research environments.

Team - 1 [Harish, Kalpana, Prakriti, Vansh]

The Challenge of Academic & Professional Research

Information Overload

Researchers and students spend countless hours reading lengthy, dense research papers, struggling to identify critical information.

Time Inefficiency

Extracting core findings, methodologies, and conclusions is a slow and tedious process that consumes valuable research time.

Context Retrieval

Finding specific answers buried in 50+ page PDFs is difficult manually, requiring researchers to re-read extensive sections.

Introducing PaperPilot

A Generative AI-powered application that acts as a virtually intelligent reading assistant for researchers and students.

Core Approach

Employs **Retrieval-Augmented Generation (RAG)** to fetch precise context from documents rather than relying on generic pre-training data.

Instant Processing

Read and process large PDF documents in seconds

Smart Summaries

Synthesize long texts into concise, actionable summaries

Structured Notes

Generate organized breakdowns with Introduction, Key Points, and Conclusion

Interactive Chatbot

Answer hyper-specific queries based entirely on provided documents

Core Technologies & Architecture

Streamlit

Frontend UI framework for a seamless, rapid Python-based web app interface

Llama 3.3

LLM engine (70b-versatile) hosted on Groq for lightning-fast, high-quality inference

LangChain

Framework to orchestrate complex GenAI pipelines, text splitting, and LLM integrations

HuggingFace

Embeddings (all-MiniLM-L6-v2) to convert text into semantic mathematical vectors

ChromaDB

Vector database for storing and retrieving high-dimensional document chunks locally

PyPDF2

Document processing library to extract raw text from PDF files

System Architecture & Workflow

Document Ingestion
User uploads a PDF document. PyPDF2 extracts all raw textual data.

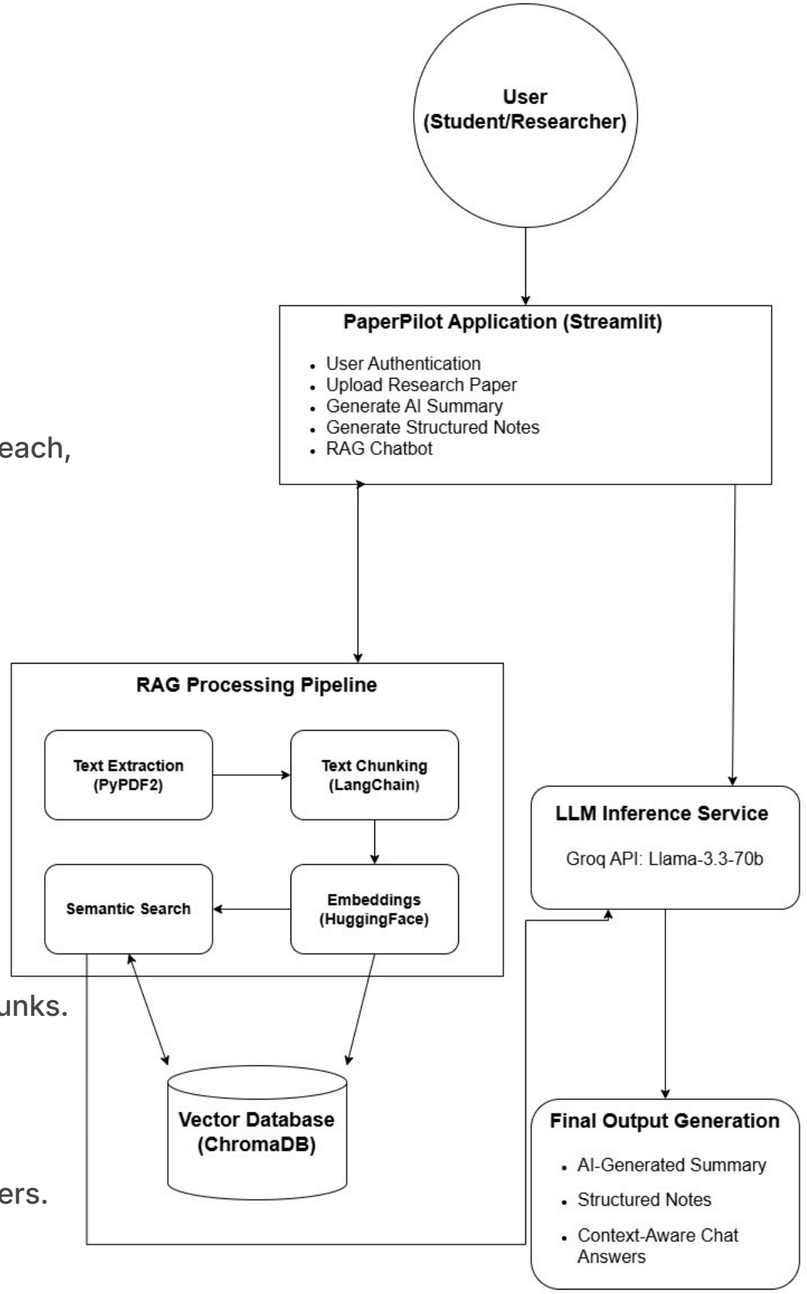
Text Chunking
RecursiveCharacterTextSplitter divides massive text into smaller overlapping chunks (1000 characters each, 200 character overlap) to maintain context.

Vector Embedding
HuggingFace converts text chunks into embeddings—semantic mathematical vector representations.

Data Storage
Vectors are saved temporarily in ChromaDB for efficient retrieval.

RAG Querying
User questions are embedded and ChromaDB performs similarity search to find top-3 most relevant chunks.

Answer Generation
Relevant chunks plus user prompt are sent to Groq/Llama-3 model, which outputs context-aware answers.



Application Features & Modules

01

Smart Authentication

Session-state managed Login/Signup system to secure user workspaces

02

Upload Module

Validates PDF format and provides interactive spinner during processing

03

AI Summary Generator

Sends first 5000 characters to generate quick, high-level abstract

04

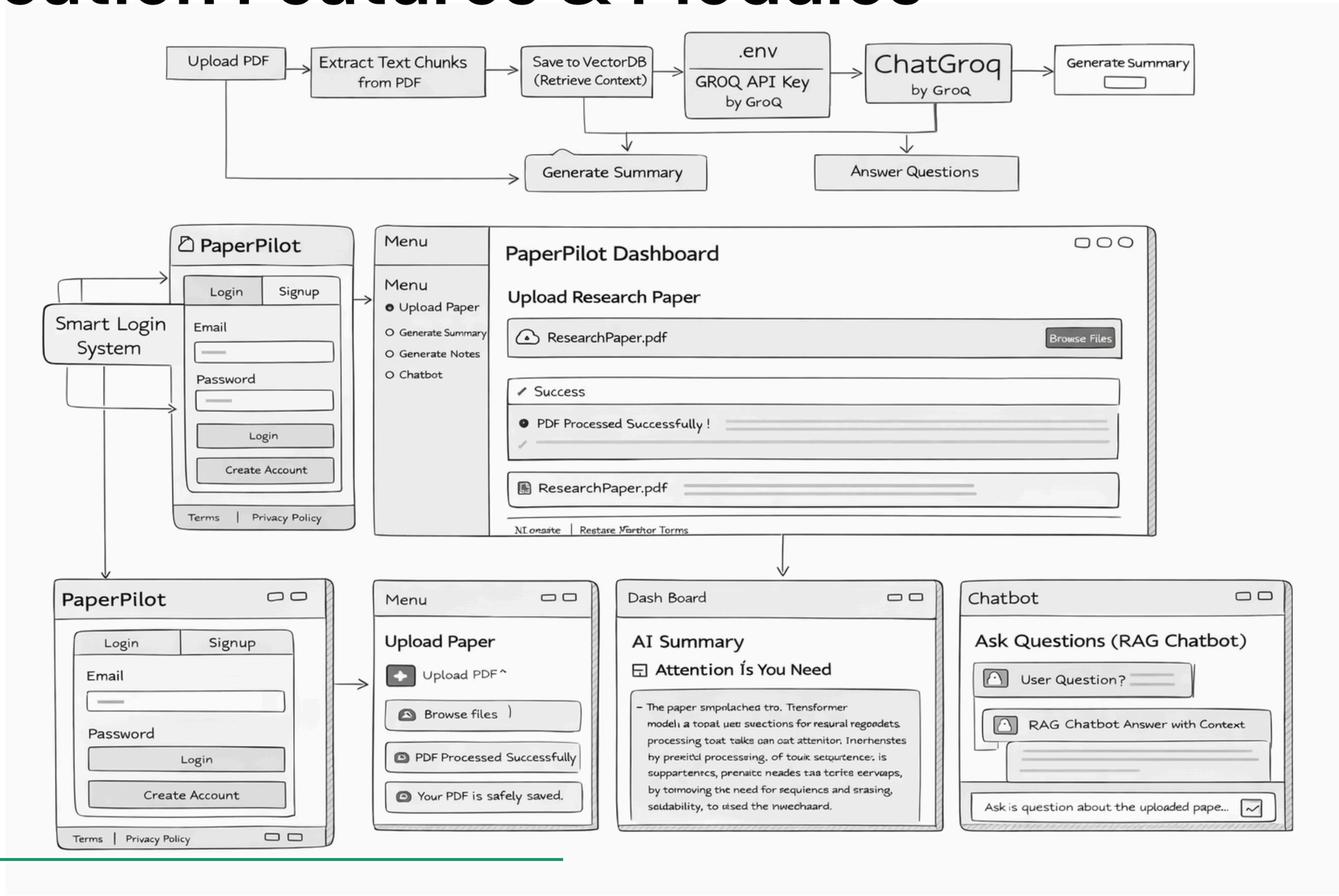
Structured Notes

Breaks down document into Introduction, Key Points, and Conclusion sections

05

Context-Aware Chatbot

Intuitive chat interface retaining conversation history with pinpointed PDF-based answers



Overcoming Implementation Challenges

Context Window Limits

Addressed using LangChain's chunking strategy to prevent sending entire 100-page PDFs to the LLM, which would crash or cost too much.

Context Loss Between Chunks

Handled by adding chunk_overlap=200 parameter during text splitting, ensuring sentences at boundaries aren't broken abruptly.

Speed vs Performance

Solved by utilizing Groq LPU inference engine combined with Llama 3 for near-instant responses rather than sluggish commercial APIs.

Secure Environment

Implemented .env for securing critical Groq API keys instead of hardcoding them in application logic.

Conclusion & Impact

Project Summary

PaperPilot successfully bridges the gap between massive academic literature and actionable insights using cutting-edge Generative AI technology.

Key Impact

Drastically reduces research time, enhances learning efficiency, and provides a scalable foundation for knowledge-retrieval applications.

"We sincerely thank the jury members and our mentors for their invaluable feedback and encouragement throughout this project."